JAVA · LTS RELEASES · INTERVIEW NOTES

Java 8, 11, 17 Features

24 features · one line each
what interviewers still ask on these three LTS lines

Java 8 Language & library

The release that made behavior something you can pass around.

1 Lambda expressions — 8

A function value for one abstract method. It can use a local only if that local is never reassigned.

```
list.forEach(x -> System.out.println(x));
```

2 Functional interfaces — 8

Exactly one abstract method. Predicate tests, Function maps, Consumer accepts, Supplier provides.

```
Predicate<String> ok = s -> s.length() > 3;
```

3 Method references — 8

A lambda that only calls an existing method: Type::static, obj::instance, Type::instance, or Type::new.

```
list.forEach(System.out::println);
```

4 Stream API — 8

filter and map are lazy. count, collect, or forEach runs the pipeline, and a stream can be used once.

```
list.stream().filter(x -> x > 0).count();
```

5 Optional — 8

A return type that is empty or holds one value. orElse always builds the fallback; orElseGet does not.

```
Optional.ofNullable(name).orElse("guest");
```

6 Default & static methods — 8

A default method is inherited and can be overridden. A static interface method is called on the interface.

```
default void log(String m) { }
```

7 java.time — 8

LocalDate has no zone. Instant is a UTC moment. ZonedDateTime has both. All three are immutable.

```
LocalDate.now().plusDays(7);
```

8 Map.computeIfAbsent — 8

Loads a missing key inside the map. If the function returns null, the key is left out.

```
cache.computeIfAbsent(id, this::load);
```

A stream does not store elements. parallel() shares the common pool, so keep it off blocking I/O.

Java 11 From 8 to 11

9 and 10 shipped in between. On an 11 codebase, these are all fair game.

1 var — 10

The compiler fixes the local's type from the initializer. Fields cannot use it, and var x = null does not compile.

```
var names = new ArrayList<String>();
```

2 List.of · Set.of · Map.of — 9

Immutable, and null elements are rejected. Set.of and Map.of also reject duplicate keys.

```
List<String> ids = List.of("a", "b");
```

3 Modules — 9

exports makes a package visible. requires names a dependency. Deep reflection needs opens.

```
module app { requires java.sql; }
```

4 String APIs — 11

strip removes Unicode spaces; trim only removes code points ≤ U+0020. isBlank is true for whitespace.

```
"  hi  ".strip();
```

5 HTTP Client — 11

java.net.http speaks HTTP/1.1 and HTTP/2. send blocks; sendAsync returns a CompletableFuture.

```
client.send(req, BodyHandlers.ofString());
```

6 Files.readString — 11

Reads a whole text file into one String, UTF-8 by default. writeString replaces that file.

```
Files.readString(Path.of("a.txt"));
```

7 Single-file launch — 11

java Hello.java compiles in memory and runs that file. It is for one file, not a multi-module build.

```
java Hello.java
```

8 Small library additions — 11

Optional.isEmpty() is the opposite of isPresent(). toArray(String[]::new) keeps the array type.

```
items.toArray(String[]::new);
```

var is still static typing. List.of returns a list you cannot add to.

Java 17 From 11 to 17

14, 15, and 16 finalized these. A Java 17 interview expects all of them.

1 Records — 16

An immutable carrier. You get a constructor, equals, hashCode, and toString. The accessor is x(), not getX().

```
record Point(int x, int y) {}
```

2 Sealed classes — 17

Only the permits list may extend it. Each permitted type is final, sealed, or non-sealed.

```
sealed interface Shape permits Circle {}
```

3 Text blocks — 15

A """ literal keeps line breaks. The newline after the opening quotes is required, and shared indent is removed.

```
var json = """
    {"ok": true}
    """;
```

4 Switch expressions — 14

Arrow labels return a value and do not fall through. Every case is covered, or there is a default.

```
switch (d) { case SUN -> 0; default -> 8; }
```

5 instanceof pattern — 16

The test and the cast are one step. The new variable is in scope only where the test is known to be true.

```
if (o instanceof String s) use(s);
```

6 Helpful NullPointerException — 14

The message names which call returned null, such as getAddress(), instead of only a line number.

```
user.getAddress().getCity();
```

7 Strong encapsulation — 17

JDK internals stay closed. Reflecting into one needs an explicit --add-opens. --illegal-access is gone.

```
--add-opens java.base/java.lang=app
```

8 Compact constructor — 16

A record body public Point { } validates arguments. Fields are assigned after that body, so you cannot read them yet.

```
public Point {
  if (y < x) throw new IllegalArgumentException();
}
```

Pair a sealed parent with records when the set of cases is closed.

How to place each LTS in an answer

JAVA 8 — Pass behavior
lambda · stream · Optional
default methods · java.time

→

JAVA 11 — Smaller platform
modules · var · List.of
HTTP client · new Strings

→

JAVA 17 — Model the data
record · sealed · text block
switch · patterns

Say which release finalized it, then the trap: orElse vs orElseGet, strip vs trim, x() not getX().

Finalized later — not Java 17

Virtual threads — Java 21	Pattern switch, final — Java 21
Record patterns — Java 21	Sequenced collections — Java 21

Pattern matching for switch was only a preview in 17.